

CHIAYI, Taiwan

WMO: 467480

Lat: 23.4959N

Lon: 120.4329E

Elev: 88

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 48.8	(c) 51.3	(d) 40.8	(e) 37.6	(f) 51.5	(g) 44.1	(h) 42.7	(i) 54.0	(j) 16.6	(k) 62.7	(l) 15.4	(m) 60.8	(n) 5.1	(o) 10	(p) 0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.7	(c) 92.5	(d) 80.4	(e) 91.4	(f) 80.4	(g) 90.4	(h) 80.1	(i) 81.9	(j) 90.0	(k) 81.3	(l) 89.3	(m) 80.7	(n) 88.6	(o) 9.7	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.7	154.5	87.5	79.0	151.0	86.5	78.4	147.7	85.7	45.9	90.4	45.1	89.6	44.5	89.0	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.0	(b) 13.3	(c) 11.8	(e) 43.3	(f) 95.1	(g) 3.5	(h) 0.7	(i) 40.8	(j) 95.6	(k) 38.7	(l) 96.0	(m) 36.7	(n) 96.4	(o) 34.1	(p) 97.0		
(4) 15.0	(5) 13.3	(6) 11.8	(7) 43.3	(8) 95.1	(9) 3.5	(10) 0.7	(11) 40.8	(12) 95.6	(13) 38.7	(14) 96.0	(15) 36.7	(16) 96.4	(17) 34.1	(18) 97.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.5	62.7	63.6	68.5	73.9	78.7	82.6	84.0	83.2	81.0	76.4	72.3	66.7
	DBStd	8.38	4.59	5.03	5.68	5.72	3.40	2.87	1.99	1.91	2.62	3.53	3.92	4.59
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	259	98	81	33	11	0	0	0	0	0	0	3	33
	CDD50	8955	395	382	575	717	889	977	1052	1030	930	820	669	519
	CDD65	3737	27	43	143	277	424	527	587	565	480	355	222	87
	CDH74	33644	80	153	657	1897	3604	5835	6954	6338	4577	2275	1011	263
(13)	CDH80	11674	5	14	126	505	1168	2321	2918	2359	1490	568	173	27
(14)	Wind	WSAvg	5.3	6.2	5.9	5.6	5.1	4.8	5.6	5.5	5.4	4.7	4.6	5.2
(15) Precipitation	PrecAvg	77.4	1.1	2.0	2.2	4.3	6.8	14.5	16.5	17.2	8.8	1.9	1.1	0.9
	PrecMax	127.2	3.5	12.9	6.3	18.0	16.0	36.1	37.8	46.0	40.4	12.9	8.3	3.7
	PrecMin	40.5	0.0	0.0	0.2	0.0	0.4	0.9	2.4	3.9	1.6	0.0	0.0	0.0
	PrecStd	22.2	0.9	2.8	1.9	4.1	3.9	8.9	9.4	12.0	9.1	2.9	1.8	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.9	82.3	86.9	89.0	91.3	94.3	93.6	92.6	91.1	88.7	86.3	83.5
		MCWB	66.5	71.1	73.1	77.5	80.6	80.1	80.8	81.5	80.7	79.0	74.4	71.7
	2%	DB	75.9	78.6	83.5	87.2	89.4	92.6	92.4	91.3	89.8	86.9	84.1	79.5
		MCWB	65.6	68.8	73.1	76.5	79.4	80.0	80.5	81.1	80.3	77.9	73.7	69.3
	5%	DB	73.3	75.4	80.4	85.4	87.8	91.1	91.4	90.2	88.6	85.2	82.0	76.8
		MCWB	64.6	67.3	71.5	75.6	78.6	80.0	80.2	80.7	79.7	76.7	72.4	67.4
	10%	DB	70.5	72.3	77.7	83.2	86.1	89.7	90.3	89.0	87.2	83.5	79.8	74.2
		MCWB	63.1	65.7	70.0	74.8	77.7	79.6	80.0	80.2	79.1	75.5	71.0	65.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.4	72.6	75.4	78.8	81.9	82.3	82.6	82.9	82.1	80.1	76.2	73.6
		MCDB	76.2	80.6	84.1	87.3	89.4	91.4	91.6	90.7	89.0	87.1	83.9	82.1
	2%	WB	66.6	70.3	73.7	77.3	80.2	81.2	81.6	81.9	81.2	78.7	74.7	70.7
		MCDB	74.3	76.7	81.5	85.7	87.8	90.2	90.4	89.6	88.1	85.1	82.2	76.7
	5%	WB	65.2	68.2	72.3	76.2	79.1	80.4	80.9	81.2	80.4	77.5	73.4	69.0
		MCDB	71.9	74.0	79.2	84.1	86.7	89.2	89.5	88.6	87.2	83.5	80.2	74.4
	10%	WB	63.7	66.0	70.8	75.1	78.0	79.7	80.2	80.5	79.6	76.4	71.9	67.4
		MCDB	69.3	72.3	76.9	82.3	85.1	87.9	88.4	87.7	85.8	81.8	78.0	72.6
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	13.5	14.4	13.8	13.4	12.6	12.7	11.8	12.9	14.6	16.0	16.4
		MCDBR	19.0	17.9	17.8	16.4	15.4	15.2	14.5	13.6	14.3	15.6	17.4	18.6
		MCWBR	10.0	9.5	9.1	8.4	7.6	6.6	6.2	6.1	6.5	7.6	8.3	9.3
	5% WB	MCDBR	17.0	16.5	17.1	15.3	14.7	14.3	13.5	13.1	13.3	13.9	15.6	15.7
		MCWBR	9.2	9.5	9.4	8.3	7.7	6.6	6.3	6.2	6.5	6.9	7.9	8.5
(40) Clear-Sky Solar Irradiance	taub		0.509	0.564	0.694	0.698	0.590	0.534	0.544	0.602	0.631	0.612	0.533	0.522
	taud		1.965	1.861	1.612	1.655	1.923	2.093	2.056	1.884	1.792	1.791	1.968	1.952
	Ebn at Noon		234	231	208	212	234	246	244	230	220	215	226	224
	Edh at Noon		53	62	83	81	62	52	54	64	69	66	52	52
(44) All-Sky Solar Radiation	RadAvg		1082	1212	1310	1398	1517	1597	1740	1570	1533	1433	1141	1052
	RadStd		117	150	139	175	188	167	127	162	122	87	93	103

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	+0.50	N/A	N/A	N/A	N/A	+0.32	+0.45	N/A	N/A	+218	+200

Nomenclature: See separate page